

A Solid-Based Web Application for Decentralized and Privacy-Preserving Energy Data Sharing in Logistics

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Abstract

Energy consumption data in logistics properties remains fragmented across organizational silos, hindering effective comparison and benchmarking. Existing centralized platforms require stakeholders to relinquish control of their data control, which conflicts with privacy requirements and commercial interests. We present the GRANERGIZE WebApp: A Solid-based application that enables decentralized sharing of building energy data while preserving data sovereignty. The application uses Web Access Control for fine-grained permissions and the Solid Application Interoperability specification for access notifications. In addition to sharing individual building data, the system supports the creation and sharing of aggregated views that only expose computed statistics without revealing underlying buildings. Our implementation demonstrates how the Solid protocol can solve real-world data-sharing issues in the logistics sector, allowing collaboration without centralized data custody.

Keywords

Solid, Linked Data, Access Control,

1. Introduction

Energy consumption data in logistics properties remains trapped in isolated silos, hindering access and exchange. The logistics ecosystem's complexity – with its diverse stakeholders and varied data formats – creates heterogeneity that prevents effective data sharing. Consequently, comparing energy consumption data across properties and organizations remains rudimentary at best.

Logistics properties form central hubs of logistics networks and therefore play a decisive role in the provision of logistics services (source: Nehm et al.). With the rapid rise in electricity and gas prices [1] and the increasing importance of sustainability and CO₂ accounting through specific EU taxonomies [2], the relevance of energy efficiency in logistics real estate is becoming ever clearer.

Yet substantial barriers persist: The stakeholder landscape's complexity – with landlords, tenants, and operators holding different perspectives on energy use – limits access to the data that is essential for efficient management. Even within organizations, mobilizing consumption data requires extensive manual effort. Furthermore, the diversity of building types and usage patterns makes meaningful comparison difficult.

Existing centralized platforms fail to address these barriers because they require stakeholders to relinquish data control and adopt singular data models. The Solid protocol, with its emphasis on decentralized data ownership and standardized access control, offers an alternative: stakeholders retain sovereignty over their data while enabling controlled sharing. This paradigm shift aligns with both GDPR principles and the commercial realities of competitive logistics markets.

To address these challenges, the GRANERGIZE project has developed an ontology that provides a unified data model for energy consumption data, as well as a WebApp for visualizing and sharing this data, which is stored on the user's Solid Pods. The WebApp provides visualizations of the energy consumption data and allows users to define access rights for real or aggregated data via an intuitive user interface. Whereas aggregated data is computed based on the user's preferences and shared as a

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snapshot without provenance information, individual building data can be shared directly from the owner's Pod with fine-grained access control.

This paper describes the Solid-based architecture

2. Scenario

3. Implementation

In the following, we describe the implementation of the GRANERGIZE WebApp. Following the Solid principles of separating identity, data, and application, the WebApp operates as a pure client-side application that communicates directly with Solid Pods without requiring a central backend. All user data – building metadata, energy measurements, and sharing configurations – resides exclusively in the users' personal Solid Pods.

3.1. System Architecture

The GRANERGIZE WebApp is implemented as a single-page application using React 18 with Vite as the build tool and Material-UI for the user interface components. Users authenticate via the Solid-OIDC protocol, which allows them to log in with their WebID and connect to the Identity Provider of their choice. All RDF processing occurs client-side using the N3.js library, which provides parsing and serialization of Turtle documents as well as an in-memory quad store for pattern matching.

3.2. Data Model

To maximize interoperability, the GRANERGIZE data model builds upon established ontologies and introduces custom terms only where necessary. For example, the data model reuses the vcard ontology for postal addresses, the RealEstateCore (REC) ontology for building classifications, and Schema.org for organizational entities. Additionally, geographic regions – cities and administrative districts – are identified using Wikidata IRIs, enabling links to external knowledge graphs. The GRANERGIZE vocabulary (gra:) defines domain-specific vocabulary for logistics buildings and energy measurements.

Static information on each building, such as: type, geographic coordinates, postal address, total area, and year of construction is stored as a separate Turtle file within the user's Pod, see <https://granergize-homer.solidcommunity.net/private/granergize/buildings/2.ttl> for an example. Additionally, the building files contain optional properties capturing domain-specific information such as photovoltaic system availability, NACE industry code, or energy certificate references. A building's energy measurement data is stored in separate datasets per year, linked to from the building file. Separating these datasets allows sharing building metadata independently of detailed energy data, supporting fine-grained access control as required by the data providers.

A building's energy data captures the full energy flow within a logistics property using a comprehensive ontology. An example file, containing the detailed energy measurements is available at <https://granergize-homer.solidcommunity.net/private/granergize/energy/2023/2.ttl>. The model distinguishes six categories: energy need, energy generation, energy storage, energy distribution, energy transfer, and energy usage. Each category contains granular measurement types, such as electricity consumption, photovoltaic generation, heat pump output.

3.3. Data Sharing and Access Control

Access control relies on the Web Access Control (WAC) specification [3], which is part of the Solid platform. WAC associates each resource with an Access Control List (ACL) stored in a corresponding .acl file. For defining access rules, ACLs provide a standardized vocabulary (acl:agent, acl:accessTo, acl:mode) to specify read-write permissions on web resources within a Solid Pod. When the data provider grants access to a building, the application appends a new acl:Authorization block

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sections
overview

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section
entirely if
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tight

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specifying the data requester's WebID and the granted access mode (typically `acl:Read`). The resource owner retains `acl:Control` permission, allowing them to modify or revoke access at any time. If the data provider chooses to share energy data as well, a separate authorization is added to the energy data file's ACL, enabling fine-grained control over which data elements are accessible.

The sharing workflow follows the Solid Application Interoperability (INTEROP) specification [4] for notifying data requesters about granted access. Figure 1 illustrates the sharing process. The data provider initiates sharing through the web interface by selecting a building and entering the recipient's WebID. Upon confirmation, the application executes three operations on the data provider's Pod (steps 1–3 in the figure): first, it updates the resource's ACL file (`building.ttl.ac1`) to grant read permission to the recipient; second, it records the sharing action in `sharingRegistry.ttl`; third, it posts an `interop:AccessGrant` notification to the recipient's inbox. The notification specifies the granter's WebID (`interop:grantedBy`), a timestamp (`interop:grantedAt`), and a data grant pointing to the shared resource.

When the data requester's application polls the inbox (steps 4–5), it discovers the new access grant and automatically adds the shared building URI to `dataSources.ttl`. From this point on, the data requester can fetch the building data directly from the data provider's Pod (step 6). The ACL on Pod A ensures that only authorized requesters – those whose WebID appears in the ACL – can successfully retrieve the resource.

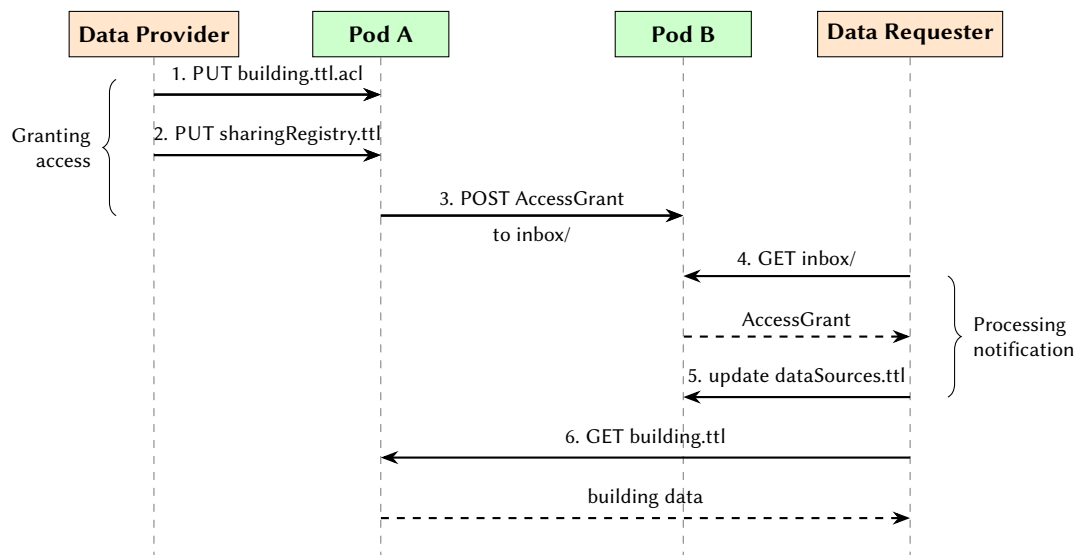


Figure 1: Sequence diagram of the data sharing process: the data provider grants access by updating the ACL and sending a notification; the data requester processes the notification and can then fetch the shared data.

The application maintains local registries to track sharing relationships on both sides. The data provider's `sharingRegistry.ttl` contains triples of the form `<buildingUri> gra:sharedWith <recipientWebId>`, enabling the WebApp to display which buildings have been shared and allowing users to easily revoke access. On the data requester's side, `dataSources.ttl` lists all external building URIs the user has been granted access to. Together, these registries ensure that both parties maintain a consistent view of their data sharing relationships without requiring a central coordination service.

Revocation mirrors the granting process by reversing each step. The application removes the recipient's `acl:Authorization` block from the resource's ACL file and updates `sharingRegistry.ttl` to delete the sharing record. On the data requester's side, subsequent access attempts return HTTP 403, prompting the application to remove the inaccessible building from `dataSources.ttl`.

3.4. Sharing Aggregated Data

While sharing individual building data enables direct collaboration, many scenarios require sharing only statistical summaries without exposing the underlying building details. For instance, a logistics provider

managing multiple warehouses may wish to share total energy consumption across their portfolio with an industry benchmarking initiative, without revealing individual building locations or identities. The GRANERGIZE WebApp addresses this requirement through aggregated views – snapshots that only contain computed values.

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An aggregated view is defined by three parameters: a set of source buildings, an aggregation function (sum, average, minimum, or maximum), and a selection of energy metrics to include. Figure 2 shows a screenshot of the user interface for creating an aggregated view. In this screenshot the user selected three buildings and the average function for gas and electricity metrics. The view definition is stored as a separate Turtle file in the owner’s Pod, as shown in Listing 1. Crucially, the view definition itself – including the source building URIs – is never shared. Only the computed snapshot, containing aggregated values without provenance information, is made accessible to recipients.

Create Aggregated View

Create an aggregated view that combines energy data from multiple buildings. The computed values will be stored as a privacy-preserving snapshot that can be shared with others without revealing the source buildings.

View Name

Average Gas & Electricity 467

Select Buildings

Building 4

Building 6

Building 7

Aggregation Type

☒ Average

☐ Sum

☐ Minimum

☐ Maximum

Metrics to Include

Energy Need

☒ gas

☒ electricity

☐ gridSupply

☐ solar

☐ solarSpaceHeating

☐ photovoltaic

☐ selfConsumption

☐ gridFeedIn

☐ hallHeatingFromWasteLoss

☐ frostProtectionHBWFromWasteLoss

☐ ambientHeat

☐ ventilationHeat

☐ personHeat

☐ groundwater

☐ woodChips

CANCEL

CREATE VIEW

```
1 <#view-1770048602886-f5n4sea> a
   ↳ gra:AggregatedViewDefinition ;
2   gra:viewId "view-1770048602886-f5n4sea" ;
3   gra:viewName "Average Gas and Electricity
   ↳ 467" ;
4   gra:aggregationType "average" ;
5   gra:createdAt
   ↳ "2026-02-02T16:10:02.886Z"^^xsd:dateTime
   ↳ ;
6   gra:includesBuilding
   ↳ <../private/granergize/buildings/4.ttl>,
7     <../private/granergize/buildings/2.ttl>,
8     <../private/granergize/buildings/6.ttl> ;
9   gra:includesMetric "gas", "electricity" ;
10  gra:lastComputedAt
   ↳ "2026-02-02T16:10:04.064Z"^^xsd:dateTime
   ↳ .
```

Listing 1: View Definition Example

Figure 2: Create Aggregated View Dialog

The aggregation workflow proceeds as follows. When the data owner creates or refreshes a view, the application fetches energy data from each source building, applies the selected aggregation function across all buildings for each metric, and stores the result as a separate Turtle file (the snapshot).

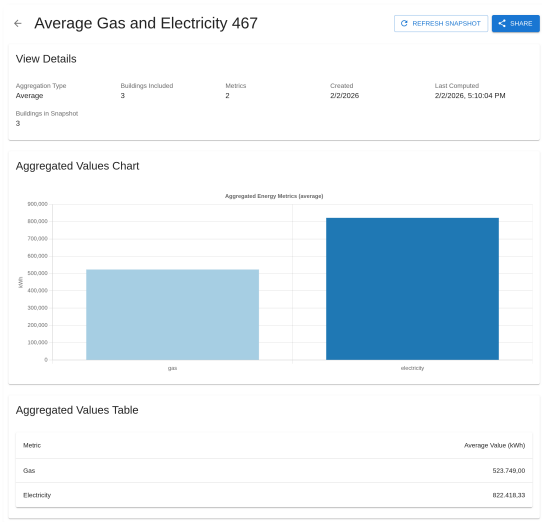


Figure 3: Create Aggregated View Dialog

```

1 <#snapshot> a gra:AggregatedViewSnapshot ;
2   gra:viewId "view-1770048602886-f5n4sea" ;
3   gra:viewName "Average Gas and Electricity
   ↪ 467" ;
4   gra:aggregationType "average" ;
5   gra:computedAt
   ↪ "2026-02-02T16:10:04.064Z"^^xsd:dateTime
   ↪ ;
6   gra:buildingCount 3 ;
7   gra:includesMetric "gas", "electricity" ;
8   gra:gasValue 523749.00 ;
9   gra:electricityValue 822418.33 .

```

Listing 2: View Definition Example

Figure 3 shows an example of such a snapshot for the view defined in Listing 1. The snapshot contains only the aggregated values and metadata such as the computation timestamp, building count, and time period – no individual building identifiers appear in the output. This materialized snapshot approach ensures that recipients cannot infer which specific buildings contributed to the aggregation.

Sharing an aggregated view follows the same access control pattern as individual buildings. The application updates the snapshot’s ACL file to grant read permission to the recipient’s WebID and posts an access notification to their inbox.

View owners retain full control over their aggregated data. Because snapshots are materialized rather than computed on-demand, the shared values reflect the state at the time of creation or last refresh. Owners can update the snapshot at any time by triggering a manual refresh, which recomputes the aggregation from current source data and overwrites the previous snapshot. Recipients automatically see the updated values on their next fetch, while the owner maintains the ability to review changes before publishing. Access revocation works identically to individual building sharing: removing the recipient’s authorization from the ACL immediately prevents further access.

4. Conclusion

Energy consumption data in the logistics sector remains fragmented across organizational boundaries, preventing effective benchmarking and collaboration. Centralized platforms require stakeholders to surrender data control, conflicting with privacy requirements and competitive interests.

This paper presented the GRANERGIZE WebApp, a Solid-based application that addresses these challenges through decentralized data sharing while preserving data sovereignty. The implementation demonstrates how the Solid protocol’s core components – Web Access Control for fine-grained permissions and the Application Interoperability specification for access notifications – can solve real-world data-sharing problems. By separating identity, data, and application concerns, the system enables collaboration without centralized data custody. The aggregated views feature extends the sharing model to support statistical summaries, allowing stakeholders to contribute to benchmarking initiatives without exposing individual building details.

Future work includes integrating automated access request workflows, data integration pipelines, and supporting additional aggregation functions such as weighted averages based on building size. We also plan to evaluate the system’s usability with logistics stakeholders in a field study to identify practical adoption barriers.

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